INTERNAL COMMISSIONING PLAYBOOK

Majo Bridge Technologies and Events

Product: **MaJo e-School Bridge** (MayaBela platform)
Other schools stay on: https://mayabela.pages.dev

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Dedicated on-premises go-live guide

Lebawi International Academy (LIA) — physical server with public IP

AUDIENCE
MayaBela engineer on site + LIA IT nominee. Written for a beginner: open this, paste this.

RESULT
LIA uses its own URL and its own database on its own machine. Fnote and other schools stay on the shared cloud, untouched.

LIA WEBSITE
`https://erp.yourdomain.com`

LIA DATABASE API
`https://api.yourdomain.com`

Golden rule — read this twice. Never type the live MayaBela cloud address `https://hwkiihonthueadbhcvfi.supabase.co` into LIA's server, LIA's website build, or LIA's Android APK. That address is the other schools' database. LIA must get a **new empty** database on LIA hardware.

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1. Who does what

Label in this guide	Who	Job
OUR SIDE	MayaBela / MaJo Bridge engineer	Install software, bind domain to the API, build LIA website, create school, test, backups, handover
LIA SIDE	LIA IT or principal's nominee	Public IP, domain login, server-room access, admin password, firewall permission, UPS, one test teacher and one test parent
BOTH	Sit together	DNS clicks, isolation test, paper handover

You do **not** log into the shared MayaBela Supabase dashboard for this job. You do **not** run `deploy-cloudflare-pages.cmd`.

2. Words you will replace everywhere

Write the real values in pencil on this page before Phase 1. Then every command below uses these names.

YOUR_PUBLIC_IP
LIA's public IPv4 (example shape 196.188.x.x)

YOURDOMAIN.COM
A domain LIA owns (example lia.edu.et)

ERP.YOURDOMAIN.COM
Website teachers and parents open

API.YOURDOMAIN.COM
LIA database API (same machine, second name)

LAN_IP
Server's inside-school address (example 192.168.1.50)

SCHOOL_ID
Short id in the app (example LIA — letters only)

Two hostnames, one IP. Teachers open the ERP name. The app talks to the API name. Both DNS records point at the same public IP.

3. Phase 0 — Collect facts (before you travel)

LIA SIDE — send this list on WhatsApp and wait. Do not travel until rows 1–4 are filled.

	Ask LIA	Write the answer
☐	Public IPv4	
☐	Is it static (does not change after reboot / overnight)?	Yes / No
☐	Domain they already own	
☐	Who has the domain login (email + 2FA phone)	
☐	Server OS today (Windows or Linux), RAM, disk, UPS	
☐	Can you sit in front of the server, or only AnyDesk?	
☐	School ID they want (example LIA)	
☐	Principal email for password-reset mail	
☐	One admin, one teacher, one parent phone for tests	

3.1 Confirm the IP is really public — OUR SIDE

1. Ask LIA to read their public IP from the ISP paper or router "WAN IP" page.
2. On **your** laptop, on a network that is **not** the school Wi-Fi (home, hotel, or mobile data), open a command prompt:

```
ping YOUR_PUBLIC_IP
```

You want replies, or at least not "destination unreachable" from a private range.

> **Stop if the IP starts with 10. / 192.168. / 172.16–31. / 100.64–127.** That is not a public IP. Parents on mobile data cannot reach it. Ask the ISP for a real static public IPv4 and port forwarding before you continue.

Minimum hardware to tell them now: 16 GB RAM (8 GB is painful), 200 GB SSD or more, always-on + UPS. Ubuntu Server 22.04 or 24.04 is the right OS. Docker-on-Windows is a last resort, not the plan.

OUR SIDE laptop checklist before you leave the office:

- MayaBela git clone on your Windows laptop
- Flutter SDK works (`flutter doctor`)
- USB stick (32 GB+) for Ubuntu installer + later file copy
- This PDF printed or on the laptop
- You will **not** deploy LIA to https://mayabela.pages.dev

4. Phase 1 — Bind the domain to the public IP

BOTH — LIA logs into the registrar; you click with them. You need two names, both pointing at the same public IP.

Hostname	Purpose
`erp.yourdomain.com`	The website (Flutter web)
`api.yourdomain.com`	The LIA database API

4.1 Open the DNS panel — exact clicks

1. Open a browser.
2. Go to wherever they bought the domain (Cloudflare, Namecheap, Google Domains, Ethio Telecom portal, or their current registrar).
3. Sign in with LIA's email. Complete 2FA if asked.
4. Click the domain `yourdomain.com`.
5. Click **DNS** or **DNS Management** or **Advanced DNS**.

4.2 Add two A records

Click **Add record** (or **Add new record**) twice.

Record 1 — website

Field	Type this
Type	A
Name / Host	`erp` If the panel already shows `.yourdomain.com` on the right, type only `erp`. If it wants the full name, type `erp.yourdomain.com`.
Value / IPv4 address	`YOUR_PUBLIC_IP`
TTL	`300` or Automatic
Proxy (Cloudflare only)	**DNS only / grey cloud**. Click the orange cloud until it turns grey. Orange proxy will break login and live updates.

Record 2 — API

Field	Type this
Type	A
Name / Host	`api`
Value / IPv4 address	the **same** `YOUR_PUBLIC_IP`
TTL	`300`
Proxy (Cloudflare only)	**DNS only / grey cloud**

Click **Save** after each record.

Optional later: a third A record Name `@` Value `YOUR_PUBLIC_IP` if they want `https://yourdomain.com` itself to open the ERP. Start with `erp` and `api` only.

4.3 Wait and check — OUR SIDE, any network

Windows: press Win+R, type `cmd`, Enter. Paste (change the names):

```
nslookup erp.yourdomain.com
nslookup api.yourdomain.com
```

You want a line **Address** = `YOUR_PUBLIC_IP` on both. If you still see old numbers, wait 5–30 minutes. Do not continue to HTTPS (Phase 7) until this is correct.

From a phone on **mobile data** (not school Wi-Fi), you will not see a website yet — that is normal until Phase 7. DNS only has to *resolve*.

5. Phase 2 — Router and firewall

LIA SIDE does the clicks; you watch. Goal: internet traffic on 80 and 443 reaches the physical server. Nothing else.

5.1 Find the server's LAN IP

If the server is already on and you can log in:

```
ip a
hostname -I
```

Write the address that looks like `192.168.x.x` or `10.x.x.x`. That is `LAN_IP`. Ask LIA IT to make a DHCP reservation so this number never changes (router page: DHCP → bind this server's MAC address to `LAN_IP`).

5.2 Port forward on the school router

Open the router admin page (often `192.168.1.1` or `192.168.0.1`). Look for **Port forwarding** / **Virtual server** / **NAT**. Add two rules:

Name	WAN port	Protocol	Forward to	LAN port
LIA-HTTP	80	TCP	`LAN_IP`	80
LIA-HTTPS	443	TCP	`LAN_IP`	443

Do not forward these: 22 (SSH) to the whole internet, 5432 (Postgres), 8000, 54321, 54322, 54323 (internal Docker). SSH stays on the LAN or a VPN.

6. Phase 3 — Install Ubuntu on the physical server

OUR SIDE on LIA hardware. If Ubuntu 22.04/24.04 is already installed, skip to Phase 4.

1. On your laptop download Ubuntu Server 24.04 LTS: **https://ubuntu.com/download/server**
2. Make a USB installer with Rufus: **https://rufus.ie** (device = the USB, boot type = the Ubuntu ISO, Start).
3. Plug the USB into the school server. Power on. Press the boot-menu key (often F12, F10, Esc, or Del). Choose the USB.

4. Install:
   - Language: English
   - Keyboard: English (US) or their layout
   - Use an entire disk **only** if they confirm this disk has no school data
   - Your name: `liaadmin` (or their IT name)
   - Server name: `lia-erp`
   - Strong password — write it in the handover envelope, not in WhatsApp
   - Install OpenSSH server = **Yes**
5. Reboot. Remove the USB. Log in on the screen as `liaadmin`.

7. Phase 4 — First Linux commands

OUR SIDE on the server. Log in. Paste **one block at a time**. Wait for the prompt to come back.

7.1 Update the machine

```
sudo apt update
sudo apt -y upgrade
sudo apt -y install ca-certificates curl git ufw nginx unzip jq
```

It will ask for the `liaadmin` password after `sudo`. Typing is invisible. That is normal.

7.2 Firewall — only the website ports

```
sudo ufw allow OpenSSH
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp
sudo ufw enable
sudo ufw status
```

When it asks “Proceed with operation (y|n)?” type `y` and Enter. You should see 80 and 443 ALLOW.

7.3 Install Docker (official)

```
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc]
https://download.docker.com/linux/ubuntu $(. /etc/os-release && echo $VERSION_CODENAME) stable" |
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
sudo apt update
sudo apt -y install docker-ce docker-ce-cli containerd.io docker-compose-plugin
sudo usermod -aG docker $USER
```

Log out (type `exit`) and log in again so Docker works without typing `sudo` every time. Then check:

```
docker --version
docker compose version
```

You want two version numbers, not “command not found”.

8. Phase 5 — Install LIA's private database stack

OUR SIDE. This is LIA's **own empty** database. It is not a copy of Fnote. It is the same *kind* of software (Postgres + Auth + Storage + Edge Functions) running on LIA hardware.

8.1 Download the official self-host bundle

```
sudo mkdir -p /opt/lia
sudo chown $USER:$USER /opt/lia
cd /opt/lia
git clone --depth 1 https://github.com/supabase/supabase.git
cd supabase/docker
cp .env.example .env
```

8.2 Edit secrets — do not skip

On your laptop (Git Bash or WSL) generate four random strings:

```
openssl rand -base64 32
openssl rand -base64 32
openssl rand -base64 32
openssl rand -hex 32
```

Write them on paper as POSTGRES_PASSWORD, a dashboard password, and extra secrets. Do **not** reuse MayaBela cloud keys.

On the server:

```
nano /opt/lia/supabase/docker/.env
```

Find and change at least these lines (use the real domain):

```
SITE_URL=https://erp.yourdomain.com
API_EXTERNAL_URL=https://api.yourdomain.com
SUPABASE_PUBLIC_URL=https://api.yourdomain.com
POSTGRES_PASSWORD=paste-a-long-random-string
```

Leave `ANON_KEY` and `JWT_SECRET` as a **matching pair** from this LIA `.env` file. For first go-live you may keep the pair that ships in `.env.example` on this isolated server, then rotate later. Never paste the **cloud** anon key here.

Save: Ctrl+O, Enter, Ctrl+X.

Copy the LIA `ANON_KEY` onto your laptop in a notepad named `LIA-ONLY-keys.txt`. You will need it in Phase 8. Do not put it in the shared cloud password manager next to Fnote keys without a clear LIA label.

8.3 Start the stack

```
cd /opt/lia/supabase/docker
docker compose pull
docker compose up -d
docker compose ps
```

Wait 1–3 minutes. Kong, Auth, Rest, Storage, and db should show **running** / healthy. If a container restarts in a loop, run `docker compose logs --tail=80` and fix the `.env` before continuing.

Kong listens on **localhost port 8000**. The internet must not hit 8000. Nginx will expose only 443.

```
curl -sS -I http://127.0.0.1:8000/auth/v1/health
```

You want an HTTP response, not "connection refused".

9. Phase 6 — Load MayaBela schema and functions

OUR SIDE. Copy the MayaBela `supabase` folder from your laptop onto this server.

9.1 Copy files from your Windows laptop

On the laptop, PowerShell (change the path and IP):

```
scp -r C:\path\to\mayabela\supabase liaadmin@YOUR_PUBLIC_IP:/opt/lia/mayabela-supabase
```

If port 22 is not open from the internet (good), copy with a USB stick instead: put the repo's `supabase` folder on USB, plug into the server, copy into `/opt/lia/mayabela-supabase`.

```
ls /opt/lia/mayabela-supabase/migrations
ls /opt/lia/mayabela-supabase/functions
```

You must see files starting `20260725120000_app_documents.sql` through the latest `insa_permission_hardening` migration, and function folders including `school-login`.

9.2 Apply every SQL migration, oldest first

Do not skip files. If one errors, stop and fix it.

```
cd /opt/lia/mayabela-supabase
ls migrations | sort
cd /opt/lia/supabase/docker
docker compose exec -T db psql -U postgres -d postgres < /opt/lia/mayabela-
supabase/migrations/20260725120000_app_documents.sql
```

Repeat the last line for **each** remaining file in `migrations/`, in filename order (the date prefix is the order). When you are more comfortable you can install Supabase CLI and run `supabase db push` against this local URL — same result.

9.3 Storage buckets

On a browser **on the school LAN only**, open:

```
http://LAN_IP:54323
```

That is Supabase Studio. Create or confirm buckets:

- `school-files` — private
- `school-branding` — public

Do not open port 54323 on the internet. After go-live, use an SSH tunnel if you need Studio.

9.4 Deploy Edge Functions

Copy MayaBela function folders into the self-host functions location used by this Docker stack (folder names must match exactly), including `_shared` and at least:

`school-login`, `school-public-brand`, `school-verify-student`, `school-change-password`, `school-upsert-account`, `school-upsert-registry`, `school-delete-account`, `school-register-parent`, `school-request-password-reset`, `school-request-student-password-reset`, `school-confirm-password-reset`, `school-refresh-claims`, `school-invite-parent`, `school-submit-application`, `platform-owner-pin`, `platform-list-schools`, `platform-create-school`, `platform-update-school`, `platform-delete-school`, `platform-upload-logo`, `platform-upload-mark`, `platform-mail-config`, `maya-assistant-chat`, `send-push`, `client-crash-report`.

Set secrets on **this** server only (not the cloud project):

- `MAIL_FROM` plus either `RESEND_API_KEY` or SMTP host/user/pass
- Platform owner PIN secret used by `platform-owner-pin`
- Optional `MAYA_AI_API_KEY` if LIA bought Maya chat

Password reset will not work until mail is set. Then:

```
cd /opt/lia/supabase/docker
docker compose restart functions
```

10. Phase 7 — HTTPS reverse proxy (the padlock)

OUR SIDE. Nginx does two jobs: ERP hostname → website files; API hostname → Kong on `127.0.0.1:8000`.

10.1 Website folder

```
sudo mkdir -p /var/www/lia-erp
sudo chown -R $USER:www-data /var/www/lia-erp
echo "LIA ERP coming soon" | sudo tee /var/www/lia-erp/index.html
```

10.2 Nginx site file

```
sudo nano /etc/nginx/sites-available/lia-erp
```

Paste this whole block. Change the two hostnames.

```
server {
    listen 80;
    server_name erp.yourdomain.com;

    root /var/www/lia-erp;
    index index.html;

    location / {
        try_files $uri $uri/ /index.html;
    }
}

server {
    listen 80;
    server_name api.yourdomain.com;

    client_max_body_size 50M;

    location / {
        proxy_pass http://127.0.0.1:8000;
        proxy_http_version 1.1;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
    }
}
```

Save: Ctrl+O, Enter, Ctrl+X. Then:

```
sudo ln -s /etc/nginx/sites-available/lia-erp /etc/nginx/sites-enabled/lia-erp
sudo nginx -t
sudo systemctl reload nginx
```

`nginx -t` must say syntax is ok. If it says “duplicate default server”, you still have the Ubuntu default site enabled — run `sudo rm /etc/nginx/sites-enabled/default` and reload again.

10.3 Free HTTPS certificates

DNS (Phase 1) must already point here.

```
sudo apt -y install certbot python3-certbot-nginx
sudo certbot --nginx -d erp.yourdomain.com -d api.yourdomain.com
```

- Email: principal or your ops email
- Agree to terms: Y
- Redirect HTTP → HTTPS: choose **2** (Yes)

Test on a phone using **mobile data** (turn school Wi-Fi off):

- Open `https://erp.yourdomain.com`
- You should see "LIA ERP coming soon"
- The padlock must be closed

> **If this fails, do not build the app yet.** Fix DNS, port 80/443 forwarding, or the grey-cloud Cloudflare setting first. Certbot cannot succeed until the public IP answers on port 80 for those two names.

11. Phase 8 — Build the LIA website so it talks only to LIA

OUR SIDE, ON YOUR LAPTOP. This is the step that protects Fnote.

> **Do not run** `deploy-cloudflare-pages.cmd`. That command updates the shared site every other school uses.

On the Windows laptop, in the MayaBela folder:

```
cd C:\path\to\mayabela
flutter pub get
flutter build web --release --pwa-strategy=none --dart-define=SUPABASE_CONFIGURED=true --dart-
define=SUPABASE_URL=https://api.yourdomain.com --dart-
define=SUPABASE_ANON_KEY=PASTE_LIA_ANON_KEY_HERE
```

Paste the LIA `ANON_KEY` from Phase 5, not the cloud key.

Check the build did not bake the cloud URL. In PowerShell:

```
findstr hwkiihonthueadbhcvfi build\web\main.dart.js
```

If that command prints any line, the build is wrong. Do not upload it. Check you did not still have a cloud URL in `.env.local`, and rebuild with the dart-defines above.

Copy files to the server (USB or scp):

```
scp -r build\web\* liaadmin@LAN_IP:/var/www/lia-erp/
```

On the server:

```
sudo chown -R www-data:www-data /var/www/lia-erp
```

Phone on mobile data: open `https://erp.yourdomain.com`. You should see the MaJo / MayaBela login, not “coming soon”. Hard refresh: Ctrl+Shift+R (Windows) or reload with cache disabled.

12. Phase 9 — Create only the LIA school

OUR SIDE in the **LIA** site, not pages.dev.

1. Open `https://erp.yourdomain.com`
2. Open Platform console (same owner-PIN flow as production, but this PIN exists only on LIA’s server)
3. Create school:
   - School ID: `LIA` (or the ID they chose)
   - Name: Lebawi International Academy
   - Modules ON = the 49 TOR modules
   - Modules OFF = these 10 ids: `campus`, `cctv`, `finance`, `hr`, `payroll`, `transport`, `transport_buses`, `transport_live_gps`, `add_driver`, `digital_ops`
4. Create the principal / school-admin account
5. Log out of platform. Log in as that admin. Change the password

Do not create Fnote or any other school on this server.

13. Phase 10 — Prove it is isolated (15 minutes)

BOTH. Tick every row before you leave.

	Test	Expected
☐	Login on https://erp.yourdomain.com as LIA admin	Works
☐	Login on https://mayabela.pages.dev as Fnote	Still works, unchanged
☐	LIA admin data appears on pages.dev	**Must not**
☐	Fnote data appears on erp.yourdomain.com	**Must not**
☐	Create a dummy student on LIA	Visible only on LIA URL
☐	Parent on mobile data opens erp.yourdomain.com	Login page, padlock closed
☐	Ask LIA to unplug WAN for 10 seconds	LIA site dies; pages.dev still lives — this is correct

If LIA data appears on the shared cloud, you used the wrong SUPABASE_URL. Rebuild Phase 8.

14. Phase 11 — Mail, backups, handover

14.1 Mail

Until mail secrets are set, parents cannot reset passwords. Configure Resend or SMTP on the LIA functions, then send one test reset to the principal.

14.2 Nightly backup — OUR SIDE, leave this running

```
sudo nano /usr/local/bin/lia-backup.sh
```

Paste:

```
#!/bin/bash
set -e
STAMP=$(date +%F)
DIR=/opt/lia/backups/$STAMP
mkdir -p "$DIR"
cd /opt/lia/supabase/docker
docker compose exec -T db pg_dump -U postgres postgres | gzip > "$DIR/postgres.sql.gz"
```

```
sudo chmod +x /usr/local/bin/lia-backup.sh
sudo crontab -e
```

Add this line, save:

```
15 2 * * * /usr/local/bin/lia-backup.sh
```

Copy `/opt/lia/backups` to a USB held by the principal every week. **Prove restore once** on a spare folder before you leave. A backup you never restored is not a backup.

14.3 Paper you leave with the school

	Item	Write it
☐	URL staff/parents use	https://erp.yourdomain.com
☐	School ID	
☐	Admin username (password in a separate envelope)	
☐	“Do not use mayabela.pages.dev for LIA”	Printed on the envelope
☐	Who reboots the server	
☐	USB backup day	
☐	MayaBela support phone	+251 911 646 444

Android APK is optional and later. If you give LIA the Fnote APK, that APK still talks to the shared cloud. A LIA phone build must use the same dart-defines as Phase 8.

When LIA’s public IP or power is down, LIA’s system is down. There is no automatic fallback to MayaBela cloud, and you should not add one if they want no cloud copy of LIA records.

15. Day-by-day order

When	Where	What
Day 0	Office	Collect IP, domain login, hardware photo. Confirm public IP is real.
Morning 1	DNS website	Add `erp` + `api` A records. Grey cloud if Cloudflare.
Morning 1	Server room	Ubuntu + Docker + firewall 80/443
Afternoon 1	Server	Self-host stack, migrations, functions, nginx, Certbot. Padlock on phone data.
Morning 2	Your laptop	Flutter web build with LIA URL only. Search the JS for the cloud project ref — must find nothing.
Morning 2	Server	Upload `build/web`, create LIA school, turn off the 10 excluded modules
Afternoon 2	On site	Isolation tests + backup restore drill + paper handover

16. What this is not

- Not copying files onto a Windows desktop and double-clicking
- Not pointing LIA at the current Supabase project
- Not updating https://mayabela.pages.dev
- Not a USB “installer” — this repo does not ship a one-click on-prem installer yet

That is why this is a commissioning visit, not a 20-minute copy. Sign the agreement as: **dedicated on-prem instance for LIA, separate from the multi-tenant cloud.**

MaJo Bridge Technologies and Events PLC · MaJo e-School Bridge / MayaBela · LIA on-prem go-live guide v1.0 · 9 November 2026 (East Africa Time).
This document is an internal install playbook. It is not the commercial TOR or the technical proposal. Shared cloud for other schools remains https://mayabela.pages.dev and must not receive LIA data.